

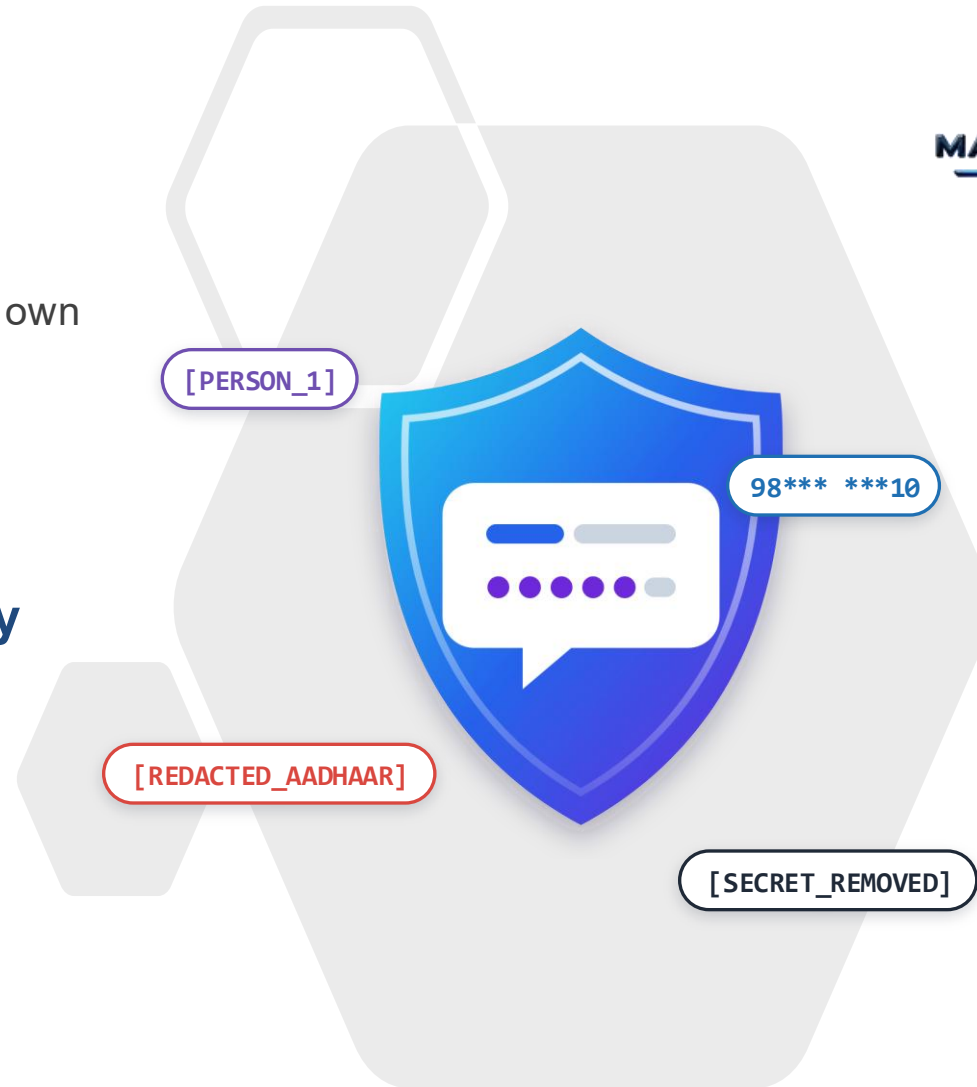
BITSHIFT 2026



P2Shield

A pre-LLM privacy firewall that cleans every prompt on your own device, before any chatbot sees it.

- Problem Statement ID – 10
- Problem Statement Title – Pre-LLM Privacy Firewall for Sensitive Data Protection
- Theme – AI / Web3
- PS Category – Software
- Team Name – TechKnights



Clean every prompt on your own device, before the LLM ever sees it

PROBLEM



77%

of GenAI users paste data into prompts (LayerX 2025)

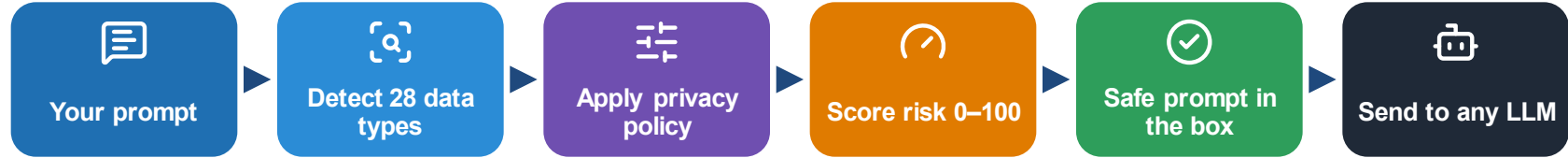
39.7%

of data shared with AI tools is sensitive (Cyberhaven 2026)

₹250 Cr

max DPDP Act penalty for failing to protect personal data

- Names, Aadhaar, card numbers, API keys and health records are pasted straight into chatbots
- Once sent, the data sits on third-party servers, outside the user's control
- Today's options: ban AI outright, or route every prompt through a cloud DLP proxy



Proposed Solution

- A **Chrome extension**: a shield appears on every AI chat box
- One click finds **28 types** of sensitive data and rewrites the prompt in place
- Mask, tokenize, redact, generalize**; secrets are always removed
- Risk score** before → after, with Undo and per-item review
- 4 policy profiles**: Personal, Healthcare, Finance, Enterprise

Innovation & Uniqueness

- 100% on-device**: no server, no account, zero network calls
- Reads **structure**: JSON, .env, headers, tables, code and prose
- Checksums** cut false alarms (Luhn, Verhoeff, IBAN mod-97)
- Optional **on-device AI** (Gemini Nano) keeps only what the task needs
- The AI only labels; the **policy decides** every replacement

How it addresses the problem – real output, Healthcare profile

BEFORE

Risk 97 · Critical

My name is **Rahul Sharma**, I'm **27**. Phone **98765 43210**, Aadhaar **2345 6789 0124**, email **rahul.sharma@gmail.com**. I have Type 2 diabetes. What should I ask my doctor?



AFTER

Risk 23 · Low

My name is **[PERSON_1]**, I'm **20-30**. Phone **[REDACTED_PHONE]**, Aadhaar **[REDACTED_AADHAAR]**, email **[REDACTED_EMAIL]**. I have **Type 2 diabetes**. What should I ask my doctor?

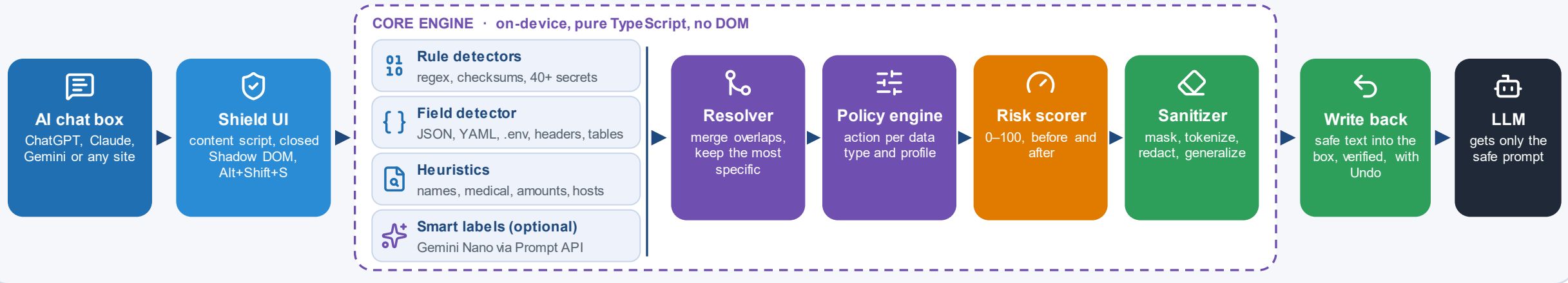
Diagnosis kept: the answer needs it. Identity removed.

TECHNICAL APPROACH

A pure TypeScript engine inside a Chrome extension. Nothing leaves the browser.



System Architecture & Process Flow



Tech Stack

TypeScript
engine + UI

Chrome MV3
extension platform

Shadow DOM
isolated shield UI

Prompt API
Gemini Nano, on-device

chrome.storage
settings, org policy

esbuild
bundler

Vitest
169 unit tests

Playwright
30 end-to-end checks

Privacy Actions

Mask

98765 43210 → 98*** **10

Tokenize

Rahul Sharma → [PERSON_1]

Redact

2345 6789 0124 → [REDACTED_AADHAAR]

Generalize

27 → 20-30 · Jamshedpur → Jharkhand

Remove secret

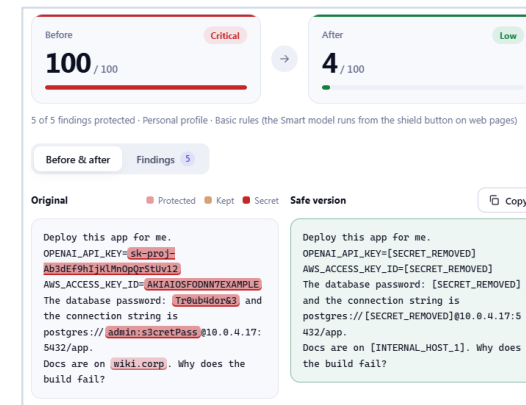
sk-proj-Ab3d... → [SECRET_REMOVED]

Keep

Type 2 diabetes (needed for the task)

Secrets are removed in every profile and cannot be overridden.

Working Prototype



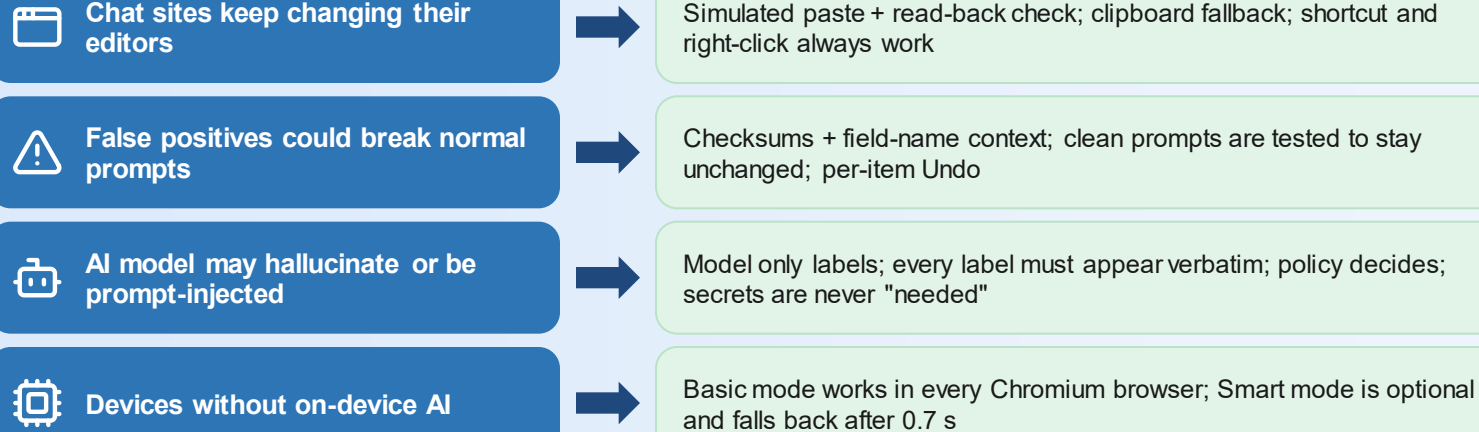
- ✓ **Shield** on every chat box
- ✓ **Alt+Shift+S** and right-click
- ✓ **Undo** + per-item review
- ✓ **Playground** for .txt .csv .json .log
- ✓ **Org policy** via managed storage

FEASIBILITY AND VIABILITY



Already built, tested and fast. No servers to run.

Risks & Mitigation



Technical Feasibility

- ✓ Working **Chrome extension**, demo chat page and playground
- ✓ **169 unit tests** + **30 end-to-end** checks, all passing
- ✓ Works on **ChatGPT, Claude, Gemini, Copilot** and more
- ✓ **No backend**: nothing to host, scale or breach
- ✓ **Org-wide policy** via Chrome managed storage

28

data types

6

actions

4

profiles

40+

secret formats

Risk score: before vs after

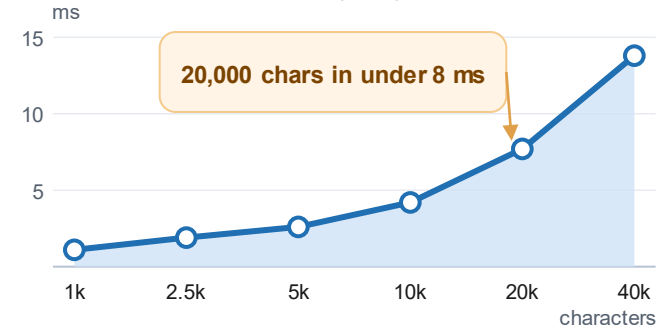
real engine output on sample prompts

Before After



Speed: analysis time vs prompt size

median of 9 runs, Basic mode, Enterprise profile



Economic Viability

- **Zero infrastructure**: runs on the user's own device
- **No per-token cost**: the on-device AI is free
- **Free** for individuals; **per-seat plan** for organizations (managed policy, audit log)
- **Free, open-source tooling**: TypeScript, esbuild, Vitest, Playwright
- DLP market: **USD 3.4 B (2025) → 13.8 B (2033)**

IMPACT AND BENEFITS

“USE AI FREELY. SHARE ONLY WHAT THE TASK NEEDS.”

WHO BENEFITS

Individuals & Students



Ask AI about health, money or studies without revealing who you are

Patients & Hospitals



Clinical facts stay, patient identity goes (Healthcare profile)

Banks & Fintech



Cards, accounts and PAN masked; salaries and amounts rounded

Developers & IT Teams



API keys, passwords and tokens stripped before debugging with AI

Enterprises & Government



Adopt AI safely instead of banning it: one policy, pushed to every browser, with a metadata-only audit log

BENEFITS



Social

Privacy by default for every AI user, so more people can use AI with confidence



Economic

Avoids breach costs (₹25.5 Cr average in India, IBM 2026) and DPDP penalties up to ₹250 Cr, with zero server spend



Compliance

Data minimisation by design (DPDP Act 2023, GDPR); the audit log stores counts, never text



Environmental

No proxy servers or extra data-centre hops; runs on devices people already have

FUTURE SCOPE



Restore real values in AI replies



PDF, DOCX and OCR for images



Hindi and other Indian languages



Local NER model for names



Org gateway for desktop apps



Tamper-evident audit log

INDUSTRY RESEARCH

77%

of GenAI users paste work data into AI prompts

[LayerX, Enterprise AI & SaaS Data Security Report 2025](#)

39.7%

of all data shared with AI tools is sensitive

[Cyberhaven Labs, AI Adoption & Risk Report 2026](#)

₹25.5 Cr

average cost of one data breach in India, a record high

[IBM, Cost of a Data Breach Report 2026](#)

2023

Samsung bans ChatGPT after engineers leak source code into it

[TechCrunch, 2 May 2023](#)

MARKET RESEARCH

Data loss prevention (DLP) market

[Grand View Research, 2026](#)

19% CAGR

2025

USD 3.4 B

2033

USD 13.8 B

About 4× growth in 8 years, driven by data-protection laws and AI adoption



Blind spot

82% of AI pastes come from personal accounts that company network tools cannot see

[LayerX 2025](#)



Gap we fill

Existing tools block AI or route prompts through a cloud proxy. P2Shield cleans them on the device, for free.



Why now

DPDP Rules 2025: full compliance due May 2027, penalties up to ₹250 Cr

[MeitY, Data Protection Framework](#)